

Archana Warriier

 archanarw.github.io

Research Interests

My research interests lie in understanding the computational principles that underlie how agents learn structured world models and use them to guide adaptive behavior, drawing on methods from probabilistic programming, causal inference, and Bayesian modeling to build AI systems with human-level capabilities.

Education

Technische Universität Kaiserslautern (RPTU)

Apr 2022 – Dec 2025

Master of Science in Computer Science

GPA: 1.6/5.0

– Relevant Coursework: Machine Learning I & II, Monte Carlo Algorithms, Automated Reasoning, Functional Programming, 3D Computer Vision, Stochastic Modeling of Cognitive Processes

Birla Institute of Technology, Mesra

June 2018 – June 2021

Bachelor of Science in Mathematics and Computing

GPA: 9.02/10

Indian Institute of Technology Madras

Jan 2021 – Aug 2021

Foundational Course on Data Science

GPA: 9.4/10

Publications

- A. Warriier, D. Nyugen, M. Naim, M. Jain, Y. Liang, K. Schroeder, C. Yang, J. B. Tenenbaum, S. Vollmer, K. Ellis, and Z. Tavares, “Benchmarking world-model learning with environment-level queries,” in *Forty-third International Conference on Machine Learning*, 2026. [Online]. Available: <https://openreview.net/forum?id=Ny6UYZ8ysN>.
- D. A. Selby, K. Spriestersbach, Y. Iwashita, D. Bappert, A. Warriier, S. Mukherjee, M. N. Asim, K. Kise, and S. J. Vollmer, “Had enough of experts? elicitation and evaluation of bayesian priors from large language models,” in *NeurIPS 2024 Workshop on Bayesian Decision-making and Uncertainty*, 2024.

Research Experience

Research Trainee

Aug 2024 – Present

Basis, New York | Advisors: Zenna Tavares, Kevin Ellis

- Developing AI systems capable of intuitive reasoning as part of Project MARA (Modeling, Abstraction, and Reasoning Agents)—a three-year initiative to build agents that can propose hypotheses, conduct experiments, and reason with abstract models of the world
- Contributing to the technical and theoretical foundations of world modeling and evaluation

Research Assistant

May 2022 – July 2024

DFKI, Kaiserslautern | Advisor: Prof. Dr. Sebastian Vollmer

- Conducted research on probabilistic inference methods and Bayesian machine learning
- Served as teaching assistant for the “Machine Learning in Julia” course

Short-term Research Scholar

May 2023 – Oct 2023

Columbia University, New York | Advisor: Zenna Tavares

- Developed a framework that automatically constructs simplified model abstractions to balance computational accuracy with resource constraints
- Extended causal abstraction theory to adapt abstractions dynamically based on query requirements and available computational resources
- Implemented proof-of-concept for adaptive abstraction in simple physics-based simulations using MuJoCo

Short-term Research Scholar

Sept 2022 – Nov 2022

Columbia University, New York | Advisor: Zenna Tavares

- Implemented parametric inversion—a program transformation technique that inverts non-injective functions by introducing parameters to uniquely determine inputs—in JAX and demonstrated the approach on simulation models, including Lotka-Volterra dynamics

- Contributed to Omega.jl probabilistic programming framework

Student Developer, Google Summer of Code

June 2021 – Aug 2021

The Julia Language Organization |

Advisors: Zenna Tavares, Sebastian Vollmer, Moritz Schauer, Jiahao Chen

- Developed CounterfactualFairness.jl package for counterfactual reasoning methods for algorithmic fairness
- Implemented multiple fairness criteria, including counterfactual fairness and path-specific effects
- Blog post: nextjournal.com/archanarw/counterfactualfairnessjl

Technical Skills

- **Programming Languages:** Julia, Python, C, Haskell, Agda, Isabelle
- **Tools:** LaTeX, Git

Teaching & Service

Teaching Assistant, “Machine Learning in Julia”

May 2022 – Aug 2024

- DFKI, Kaiserslautern | Assisted students with probabilistic programming concepts and Julia implementation

Content and Publicity Head, IEEE Student Branch

2020 – 2021

- Birla Institute of Technology, Mesra | Conducted workshops and organized technical events

Mentor, MTTS Program

2020

- Supported students in the Level 0 advanced mathematics program

Selected Honors

- Top 300, National Programming and Data Science Qualifier, IIT Madras (95% score) *2020*
- Letter of appreciation from Union Minister of India, Smriti Irani *2015*
- Bronze Medal, International Olympiad of Mathematics *2013*

Open Source Contributions

- **Active Contributor:** MARAProtocol, Autumn.cpp, Omega.jl, Jaxy.jl
- **Maintainer:** CounterfactualFairness.jl, NormalizingFlows.jl